

RVE settings

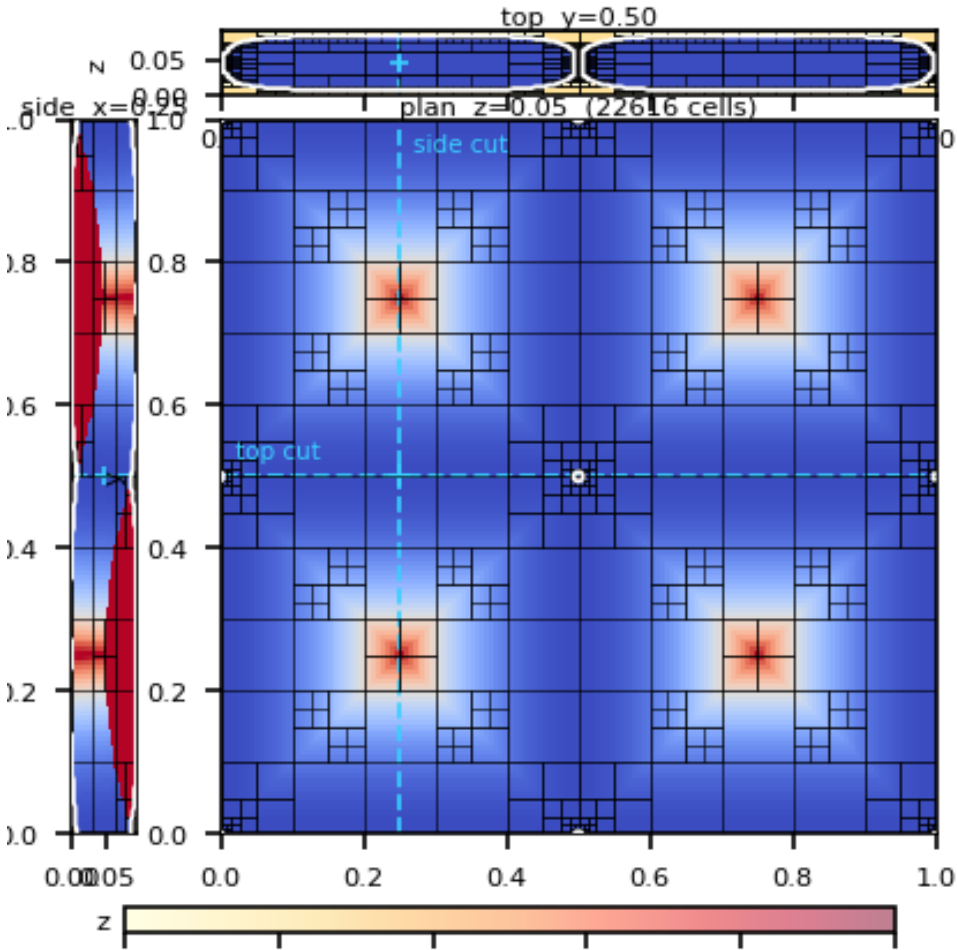
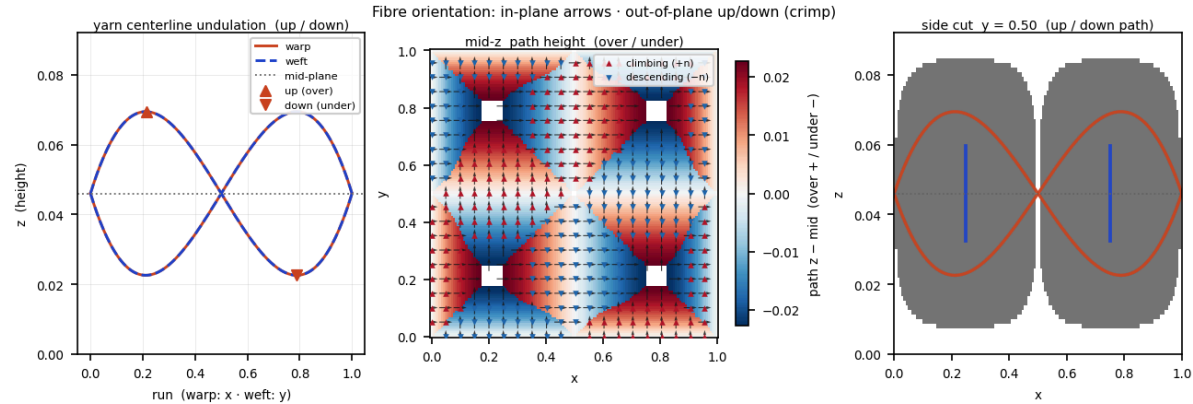
RVE size [m]	1 x 1 x 0.092
matrix material	matrix
yarn material	yarn
field type	woven
pattern	plain
n_warp	2
n_weft	2
warp width	0.49
warp height	0.076
section power	4.0
compaction	0.4
nominal Vf	0.55
max Vf	0.9
crossover nesting	on

Micromechanics

matrix (matrix)	E = 3.00 GPa, nu = 0.35
fibre (fibre)	E L = 230.00, E T = 15.00, G LT = 15.00, nu LT = 0.20
yarn model	chamis
yarn Vf range	nominal 0.55, max 0.90
Vf_b bundle/RVE (MC)	0.931
Vf_local in-tow	0.55–0.90 (μ =0.69)
Vf_avg RVE fibre	0.647

Analysis

backend	mfem-periodic
cell type	hexahedron
homogenization mesh	40 x 40 x 8
BC	periodic (3-axis MPC / saddle-point)
material sampling	local_cloud
sampling resolution	6
periodic tolerance	1e-08
homogenization AMR	on
AMR iterations	2
AMR threshold	0.2
AMR dof budget	200000
AMR figure (right)	on — base 10×10×3 hex, 4 pass(es), τ =0.2; cuts plan z =0.05, top y =0.50, side x =0.25



Engineering constants $E = (58.5, 58.5, 8.5)$ GPa; $G = (3.87, 3.30, 3.30)$; $\nu = (0.09, 0.36, 0.36)$

C_{eff} [GPa]

11	60.29	6.54	3.50	0.00	0.00	0.00
22	6.54	60.29	3.50	0.00	0.00	0.00
33	3.50	3.50	8.83	0.00	0.00	0.00
23	0.00	0.00	0.00	3.30	0.00	0.00
13	0.00	0.00	0.00	0.00	3.30	0.00
12	0.00	0.00	0.00	0.00	0.00	3.87